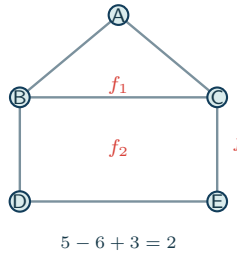
GRAPH THEORY LADDER 6 OF 9

Planar graphs and Euler's formula

A graph drawn in the plane without crossings satisfies $V - E + F = 2$, which limits how many edges it can have.

1 / THE RULE AND THE PICTURE

Vertices minus edges plus faces equals two for a connected planar graph



$5 - 6 + 3 = 2$



K_4: $4 - 6 + 4 = 2$

A face is a region of the plane cut out by the drawing, counting the outside, and for a connected graph drawn without crossings

$$\boxed{V - E + F = 2}$$

The house has 5 vertices, 6 edges and 3 faces (the roof, the room and the outside), and $5 - 6 + 3 = 2$.

2 / WHY IT HOLDS

Build the graph one edge at a time

A tree has $E = V - 1$ and a single face, the outside, so $V - E + F = V - (V - 1) + 1 = 2$. Adding an edge that closes a cycle splits one face into two, raising E and F by 1 each, and adding an edge to a new vertex raises V and E by 1 each, so $V - E + F$ never changes from 2.

3 / HOW MANY EDGES A PLANAR GRAPH CAN HAVE

At most $3V - 6$, and at most $2V - 4$ without triangles

Each face is bounded by at least 3 edges and each edge borders two faces, so $3F \le 2E$, and combining with $V - E + F = 2$ gives $E \le 3V - 6$. The graph K_5 has $10 > 3 \cdot 5 - 6 = 9$ edges, so it cannot be drawn without crossings, and $K_{3,3}$ has no triangles, so each face has at least 4 edges, giving $E \le 2V - 4 = 8 < 9$, which rules it out too.
Kuratowski showed that a graph is planar exactly when it contains no subdivision of K_5 or $K_{3,3}$.

4 / THE FIVE REGULAR SOLIDS

Every one of them gives 2

solid	V	E	F	$V - E + F$
tetrahedron	4	6	4	2
cube	8	12	6	2
octahedron	6	12	8	2
dodecahedron	20	30	12	2
icosahedron	12	30	20	2

A polyhedron is a planar graph, drawn by flattening it, so the formula holds for each of them.

5 / ON OTHER SURFACES

A doughnut changes the answer to 0

Drawn on a torus without crossings, with faces that are discs, a graph satisfies $V - E + F = 0$, and both K_5 and $K_{3,3}$ can be drawn that way. The complete graph K_7 fits on the torus with 14 triangular faces, giving $7 - 21 + 14 = 0$.

6 / TRY THESE

Use the inequalities

(a) A connected planar graph has $V = 6$ and $E = 9$: how many faces? $F = 2 - 6 + 9 = 5$.
(b) Show that the Petersen graph is not planar: it has $V = 10$, $E = 15$ and girth 5, so $F = 7$ but $5F \le 2E = 30$ gives $F \le 6$, a contradiction.
(c) What is the largest E for a planar graph on 8 vertices? $3 \cdot 8 - 6 = 18$.

ASIDE / FACES AND CYCLES

The count $E - V + 1$ of independent cycles from the tree sheet equals $F - 1$, the number of inner faces, which is the same formula read twice.

LINKS TO OTHER SHEETS

The next sheet colours the faces and vertices of planar graphs, where the four colour theorem lives.

Convention. V vertices, E edges, F faces including the outside; the bound $E \le 3V - 6$ needs $V \ge 3$; $K_{3,3}$ has no triangles.